

User Information & Communication Commands

W

Definition

The `w` command displays information about **currently logged-in users** and what they are doing.

It shows:

- Who is logged in.
- Their terminal.
- Login time.
- Idle time.
- CPU usage.
- Current running command.



Syntax

W

Example

W

Example Output

```
18:35:21 up 2 days,  3:14,  2 users,  load average: 0.15,
0.20, 0.18
```

USER	TTY	FROM	LOGIN@	IDLE	JCPU	PCPU
WHAT						
parimal	pts/0	192.168.1.15	09:12	3:00	0.10s	
0.02s	bash					
root	pts/1	localhost	10:45	0.00s	0.50s	
0.30s	top					



Explanation

- **USER** → Username.
- **TTY** → Terminal used.
- **FROM** → Remote host/IP.
- **LOGIN@** → Login time.
- **IDLE** → Idle duration.
- **JCPU** → CPU time used by all processes attached to the terminal.
- **PCPU** → CPU time of the current process.
- **WHAT** → Current command being executed.



Use Cases

- Monitor logged-in users.
- Check whether users are active.
- Troubleshoot high CPU usage.
- System administration.

Notes

- Shows much more information than who.
- Requires no special permissions.

who

Definition

Displays information about users currently logged into the system.

Syntax

who

Example

who

Example Output

```
parimal pts/0 2026-07-29 09:15 (192.168.1.15)
root pts/1 2026-07-29 10:02 (:0)
```

Explanation

Shows:

- Username
- Terminal
- Login date and time
- Remote hostname/IP (if applicable)

Use Cases

- Check who is logged in.
- Verify remote SSH users.

Notes

- Simpler than w.
- Does not show what users are doing.

users

Definition

Displays the usernames of users currently logged in.

Syntax

users

Example

users

Example Output

```
parimal root
```

Explanation

Prints only usernames separated by spaces.

Use Cases

- Quickly check logged-in users.
- Useful in shell scripts.

Notes

- No login time or terminal information.
- Very lightweight command.

last

Definition

Displays the login history of users by reading the `/var/log/wtmp` log file.

Syntax

last

Example

last

Example Output

```
parimal pts/0 192.168.1.15 Wed Jul 29 09:10 still logged in
root    pts/1 localhost    Tue Jul 28 18:20 - 18:45 (00:25)
```

Explanation

Shows:

- Username
- Terminal
- Login source
- Login time
- Logout time
- Session duration

Use Cases

- Audit login history.
- Investigate unauthorized logins.
- Track user sessions.

Notes

- Reads data from `/var/log/wtmp`.
- Requires that login history is being recorded.

id

Definition

Displays the user and group identity of a user.



Syntax

id

Or

id username



Example

id

Example Output

```
uid=1000(parimal) gid=1000(parimal)
groups=1000(parimal),27(sudo),998(docker)
```



Explanation

- **UID** → User ID.
- **GID** → Primary Group ID.
- **Groups** → All groups the user belongs to.



Use Cases

- Verify user identity.
- Check group memberships.
- Troubleshoot permission issues.



Notes

- Very useful for Linux permissions.
- Supports querying another user:

```
id root
```

```
write
```



Definition

Sends a message to another logged-in user's terminal.



Syntax

```
write username terminal
```



Example

```
write parimal pts/0
```

Type your message:

Hello!

Meeting in 5 minutes.

Press:

Ctrl + D

to finish.

Explanation

The recipient immediately sees your message on their terminal.

Use Cases

- Notify another user.
- Send quick messages on multi-user systems.

Notes

- Both users must be logged in.
- The recipient can disable messages using:

```
mesg n
```

```
wall
```

Definition

Broadcasts a message to **all logged-in users**.
("wall" stands for **write all**.)

Syntax

```
wall
```

```
or
```

```
wall "System will reboot in 10 minutes."
```

Example

```
wall "Server maintenance starts in 5 minutes."
```

Explanation

Every logged-in user receives the message on their terminal.

Use Cases

- Maintenance announcements.
- Emergency notifications.
- Administrator broadcasts.

Notes

- Usually used by system administrators.
- Very useful on multi-user servers.

```
chmod +t directory
```

Definition

Sets the **Sticky Bit** on a directory.

The sticky bit allows **everyone to create files** in the directory, but **only the file owner, the directory owner, or root can delete or rename those files**.



Syntax

```
chmod +t directory_name
```



Example

```
chmod +t /shared
```

or

```
chmod 1777 /shared
```



Explanation

Without Sticky Bit:

- Any user with write permission on the directory can delete any file.

With Sticky Bit:

- Users can delete **only their own files**.
- The directory owner or root can still delete any file.

Example permissions:

```
drwxrwxrwt
```

Notice the **t** at the end.



Common Example

The /tmp directory:

```
ls -ld /tmp
```

Example output:

```
drwxrwxrwt 10 root root 4096 Jul 29 10:00 /tmp
```

Everyone can create temporary files, but users cannot delete each other's files.



Use Cases

- Shared directories.
- Team collaboration folders.
- Public upload directories.
- Temporary storage (/tmp).



Notes

- The sticky bit is **primarily meaningful on directories**.
- On modern Linux systems, setting the sticky bit on regular files generally has **no effect**.
- Numeric equivalent:

```
chmod 1777 directory_name
```

u



Definition

There is **no standard Linux command named u** on most distributions.



Check What It Is

```
type u
```

or

```
which u  
or  
alias u
```

Explanation

If `u` works on your system, it is most likely:

- A shell alias.
- A shell function.
- A custom script.
- A command installed by a specific tool.

Example

Some users define:

```
alias u='sudo apt update'
```

or

```
alias u='cd ..'
```

Notes

- `u` is **not part of standard Linux**.
- Its behavior depends entirely on your system configuration.